6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 8

Fair division (part II)

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Indivisible items

- One or more items to be divided
  - paintings, jewellery, house, car
- n agents who want these goods
- **Goal:** divide items among agents as fairly as possible

Applications

Allocation of indivisible goods appears in many real-world scenarios:

- dividing inheritance items among heirs,
- allocating tasks among team members,
- assigning dormitory rooms to students, and
- distributing computing resources in data centers.

The fundamental challenge is that we often cannot achieve perfect fairness, so we need to explore approximations that maintain some fairness properties.

Model

An **indivisible goods allocation problem** involves:

- A set G of m indivisible goods
- A set of players $N = \{1, \ldots, n\}$
- Valuations $v_i : 2^G \rightarrow R$ (for computational purposes, Z) for each player, where $v_i(\emptyset) = 0$
  - Valuations are *additive* if for all $S \subseteq G$ and $i \in N$, $v_i(S) = \sum_{g \in S} v_i(g)$
- An allocation is a partition of the goods, denoted $A = (A_1, \ldots, A_n)$
  - $A_i \subseteq G$ is the bundle allocated to agent i
  - $A_i \cap A_j = \emptyset$ for $i \neq j$
  - allocation is *complete* if $\cup_{i \in N} A_i = G$ (and *partial* otherwise)

Basic solution concepts

An allocation $A = (A_1, \ldots, A_n)$ is

- **proportional** if $v_i(A_i) \geq v_i(G)/n$ for all $i \in N$
- **envy-free** $v_i(A_i) \geq v_i(A_j)$ for all $i, j \in N$
- **equitable** $v_i(A_i) = v_j(A_j)$ for all $i, j \in N$
- **Pareto-optimal** if there is no other allocation $B = (B_1, \ldots, B_n)$ such that $v_i(B_i) \geq v_i(A_i)$ for all $i \in N$ and $v_i(B_i) > v_i(A_i)$ for some $i \in N$

Basic solution concepts

- The core challenge with indivisible goods is that envy-freeness and proportionality are generally infeasible.
- Unlike divisible settings (like cake cutting), we cannot simply divide each good to ensure everyone gets their proportional share.
- There are instances that do not admit proportional, complete envy-free or equitable allocations.
  - 2 agents, 1 item
  - both assign positive value to the item

Relaxations: PROP1 and EF1

Definition

An allocation $A = (A_1, \ldots, A_n)$ is **proportional up to 1 good** (PROP1) if for each agent i there is an item $g \in G$ such that $v_i(A_i \cup \{g\}) \geq v_i(G)/n$.

Definition

An allocation $A = (A_1, \ldots, A_n)$ is **envy-free up to 1 good** (EF1) if for each pair of agents i, j there is an item $g \in A_j$ such that $v_i(A_i) \geq v_i(A_j \setminus \{g\})$.

- 2 agents, 1 house:
  - giving the house to one agent is PROP1 and EF1

PROP1 vs. EF1

Does EF1+completeness imply PROP1?

- NO for *general* valuations
  - $n > 2$ items, n agents, $v_i(G) = 1$, $v_i(T) = 0$ for all i and all $T \neq G$
  - giving each agent one item is EF (and hence EF1) and complete, but not PROP1
- YES for *additive* valuations
  - suppose $A = (A_1, \ldots, A_n)$ is EF1 and complete
  - consider agent i; let g^* be her most valuable item in $G \setminus A_i$
  - for every other agent j, by EF1 we have $v_i(A_i) \geq v_i(A_j \setminus \{g\}) = v_i(A_j) - v_i(g)$ for some $g \in A_j$
  - $v_i(g) \leq v_i(g^*)$, so $v_i(A_i \cup \{g^*\}) = v_i(A_i) + v_i(g^*) \geq v_i(A_j)$
  - thus, $n \times v_i(A_i \cup \{g^*\}) \geq \sum_{j \in N} v_i(A_j) = v_i(G)$

EF1: Round-robin

Round-robin (RR) algorithm:

- agents take turns choosing their most preferred item among the remaining items
- agent i chooses in round $i, n+i, 2n+i, \ldots$

	g_1	g_2	g_3	g_4	g_5
a_1	20	**50**	12	**45**	3
a_2	35	22	**90**	40	**4**
a_3	**30**	80	7	18	25

EF1: Round-robin

Theorem

For additive valuations, round-robin outputs an EF1 (and hence PROP1) allocation.

Proof.

- consider agents i and j, $i < j$
- i does not envy j:
  - i's first item is no worse to i than j's first item,
  - i's second item is no worse to i than j's second item, $\ldots$
- if we remove i's first item, j does not envy i:
  - j's first item is no worse to j than i's second item,
  - j's second item is no worse to j than i's third item, $\ldots$

$\square$

Round-robin: Pareto-optimality

Round-robin is not Pareto-optimal, even for additive valuations:

- 2 agents, 4 items
- agent 1's valuations: 10, 9, 8, 7
- agent 2's valuations: 100, 3, 2, 1
- RR:
  - items 1 and 3 to agent 1,
  - items 2 and 4 to agent 2
- Improving trade:
  agent 1 exchanges item 1 for items 2 and 4

Round-robin: strategyproofness

Round-robin is not strategyproof, even for additive valuations:

- 2 agents, 4 items
- agent 1's valuations: 9, 8, 2, 1
- agent 2's valuations: 1, 9, 2, 3
- RR for truthful agents:
  - items 1 and 3 to agent 1 (total utility: 11),
  - items 2 and 4 to agent 2
- But what if agent 1 picks item 2 first?
  - agent 2 will then take item 4
  - agent 1 can now take item 1 (total utility: 17)

EF1: beyond additive valuations

- Round-robin is not even defined for non-additive valuations
- Can we achieve EF1 for *monotone* valuations?
  - for all $S \subseteq T \subseteq G$ and $i \in N$, $v_i(S) \leq v_i(T)$
- Consider a partial allocation that is EF1.
  How can we extend it?
  - suppose there is an agent i such that no one envies i
  - if we add an unallocated item to i's bundle, the allocation remains EF1
  - but what if each agent is envied by someone?

Envy graph

- Vertices: agents
- There is an arc from i to j if i envies j
- Can be constructed in polynomial time

	g_1	g_2	g_3	g_4	g_5
a_1	**20**	50	12	45	**3**
a_2	25	**22**	**10**	40	4
a_3	30	10	7	**18**	25

Idea: reallocate along a cycle (RAC)

- Suppose the envy graph contains a cycle $i \to j \to k \to \ldots \to t \to i$
- We can reallocate bundles along the cycle
- Reallocation along the cycle (RAC):
  - give A_j to i
  - give A_k to j
  - $\ldots$
  - give A_i to t
- The set of bundles does not change, so the allocation remains EF1
  - if i now envies j by more than one item, then before RAC it envied the previous holder of j's current bundle at least as much
- Does RAC eliminate the cycle?
  - it can be shown that it eliminates an envy edge $\ldots$

Bounding the number of RAC steps

- An RAC step does not change the set of bundles
- Let all agents rank the current n bundles, breaking ties arbitrarily
- An agent's *score* for a bundle: n - rank
- Score of an allocation:
  sum of agents' scores for their bundles
- An RAC step:
  - makes all agents on the cycle strictly happier
  - does not affect the allocation of other agents
  - hence, increases the total score
- As the total score can range from 0 to $n(n-1)$, there can be at most $n(n-1)$ consecutive RAC steps

Putting it all together: envy-cycle elimination (ECE) algorithm

- EF1 allocation of goods (general monotone valuations):
  - start with an empty allocation A (EF1)
  - repeat until all items are allocated:
    - if there is an unenvied agent, allocate them an item that has not yet been allocated (A remains EF1)
    - otherwise, there is an envy cycle
    - perform an RAC step (A remains EF1)
- Correctness: if no RAC step can be performed, there is no cycle, hence there is an unenvied agent
- Polynomial runtime:
  - m item allocation steps
  - $\leq n(n-1)$ RAC steps between any two allocation steps

Envy-cycle elimination: Pareto-optimality

Envy cycle elimination algorithm is not Pareto-optimal, even for additive valuations:

- 2 agents, 4 items
- agent 1's valuations: 100, 0, 1, 5
- agent 2's valuations: 0, 100, 5, 1
- Envy cycle elimination algorithm (for some way of breaking ties):
  - item 1 to agent 1,
  - item 2 to agent 2
  - item 3 to agent 1
  - item 4 to agent 2
- Improving trade: swap items 3 and 4

Fairness and efficiency

- EF: Does not always exist
- EF-1: Always exists for additive (Round-robin) and monotone (ECE) valuations
- Cannot satisfy EF1 along with Pareto Optimality (PO) in the general case

Concepts of welfare

- **Utilitarian** welfare: sum of utilities
  $UW(A) = \sum_{i \in N} v_i(A_i)$
  - all items to a_1
- **Egalitarian** welfare: minimal utility
  $EW(A) = \min_{i \in N} v_i(A_i)$
  - 3 items to a_2, one item to a_1 and one item to a_3
- **Nash** welfare: product of utilities
  $NW(A) = \Pi_{i \in N} v_i(A_i)$
  - 2 agents get two items each, 1 agent gets 1 item

	g_1	g_2	g_3	g_4	g_5
a_1	10	10	10	10	10
a_2	1	1	1	1	1
a_3	3	3	3	3	3

Concepts of welfare

- **Utilitarian** welfare
  - always Pareto-efficient but may be highly unfair
- **Egalitarian** welfare
  - neither PO nor EF1
- **Nash** welfare
  - both PO and EF1 under additive valuations

Envy-Free up to Any Good (EFX)

Definition

An allocation $A = (A_1, \ldots, A_n)$ is **envy-free up to 1 good** (EF1) if for each pair of agents i, j <u>there is</u> an item $g \in A_j$ such that $v_i(A_i) \geq v_i(A_j \setminus \{g\})$.

Definition

An allocation $A = (A_1, \ldots, A_n)$ is **envy-free up to any good** (EFX) if for each pair of agents i, j and <u>for each</u> item $g \in A_j$ we have $v_i(A_i) \geq v_i(A_j \setminus \{g\})$.

- EFX is strictly stronger than EF1 and strictly weaker than perfect envy-freeness (EF).
- The key difference from EF1 is that EFX requires no envy after removing any good from the envied bundle, not just a carefully chosen one.

EFX: research frontier

EFX is currently one of the most intriguing open problems in fair division. We know:

- An EFX allocation exists for two players with monotonic valuations.
- An EFX allocation exists for three players with additive valuations.
- For $n \geq 4$ players with additive valuations, the existence of EFX allocations remains an open problem.

EFX for identical additive valuations

Theorem

EFX allocations are guaranteed to exist (and can be found in polynomial time) if all agents have identical additive valuations.

Proof.

- order items from largest to smallest
- allocate one by one, giving each item to a least satisfied agent
- suppose that, at the end, i envies j
- let g be the last item received by j (this is j's smallest item)
- when j received it, i's utility was at least as high as j's
- hence, $v_i(A_i) \geq v_i(A_j \setminus \{g\})$

$\square$

THANK YOU!